

Roles

Procedure: Attaching a Role to an EC2 Instance

The screenshot displays the AWS IAM console interface. At the top, the browser address bar shows the URL: `848186563482-bfpxyo37.us-east-1.console.aws.amazon.com/iam/home#/home`. The AWS logo and navigation icons are visible in the top left. The main header includes a search bar and the user profile for Archana (8481-8656-3482) with the role 'root'.

The left sidebar contains the navigation menu for IAM, including 'Dashboard', 'Access Management' (with sub-items like User groups, Users, Roles, Policies, Identity providers, Account settings, Root access management, and Temporary delegation requests), and 'Access reports' (with sub-items like Access Analyzer and Resource analysis).

The main content area is titled 'IAM Dashboard' and features several key sections:

- New! IAM Outbound Identity Federation:** A blue banner announcing the use of short-lived JSON Web Tokens (JWTs) for authentication, with an 'Enable feature' button.
- Security recommendations:** A section showing two recommendations with green checkmarks:
 - Root user has MFA: Having multi-factor authentication (MFA) for the root user improves security for this account.
 - Root user has no active access keys: Using access keys attached to an IAM user instead of the root user improves security.
- IAM resources:** A section titled 'Resources in this AWS Account' with a table listing various IAM resources. The table has columns for 'User groups', 'Users', 'Roles', 'Policies', and 'Identity providers'.
- AWS Account:** A section displaying account details:
 - Account ID: 848186563482
 - Account Alias: Create
 - Sign-in URL for IAM users in this account: `https://848186563482.signin.aws.amazon.com/console`
- Quick Links:** A section with a link to 'My security credentials'.

The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 10:45 AM on 4/7/2026.

Create the Role: Go to IAM > Roles > Create role

The screenshot displays the AWS IAM console interface. The left-hand navigation pane is expanded to show 'Identity and Access Management (IAM)', with 'Roles' selected under the 'Access Management' section. The main content area is titled 'Roles (96)' and includes a search bar and a table of existing roles. The table has columns for 'Role name', 'Trusted entities', and 'Last activity'. Several roles are listed, including '23-03-26', '26-03-26', 'AmazonSageMaker-ExecutionRole-20230505T100707', 'aws-elasticbeanstalk-service-role', 'awsrek-role-kwvnue3a', 'AWSServiceRoleForAmazonElasticFileSystem', 'AWSServiceRoleForApplicationAutoScaling_DynamoDBTable', and 'AWSServiceRoleForBackup'. The 'Last activity' column shows various timestamps, with some roles marked as '997 days ago' or '706 days ago'. At the top right of the console, there are buttons for 'Delete' and 'Create role'. The bottom of the image shows the Windows taskbar with the search bar and several application icons.

848186563482-bfpxyo37.us-east-1.console.aws.amazon.com/iam/home#/roles

aws IAM > Roles

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ **Access Management**

- User groups
- Users
- Roles**
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

▼ **Access reports**

- Access Analyzer
- Resource analysis

Roles (96) Info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Search

☐	Role name	Trusted entities	Last activity
☐	23-03-26	AWS Service: ec2	-
☐	26-03-26	AWS Service: ec2	11 days ago
☐	AmazonSageMaker-ExecutionRole-20230505T100707	AWS Service: sagemaker	997 days ago
☐	aws-elasticbeanstalk-service-role	AWS Service: elasticbeanstalk	-
☐	awsrek-role-kwvnue3a	AWS Service: lambda	-
☐	AWSServiceRoleForAmazonElasticFileSystem	AWS Service: elasticfilesystem (Servi	106 days ago
☐	AWSServiceRoleForApplicationAutoScaling_DynamoDBTable	AWS Service: dynamodb.application	706 days ago
☐	AWSServiceRoleForBackup	AWS Service: backup (Service-Linked	21 hours ago

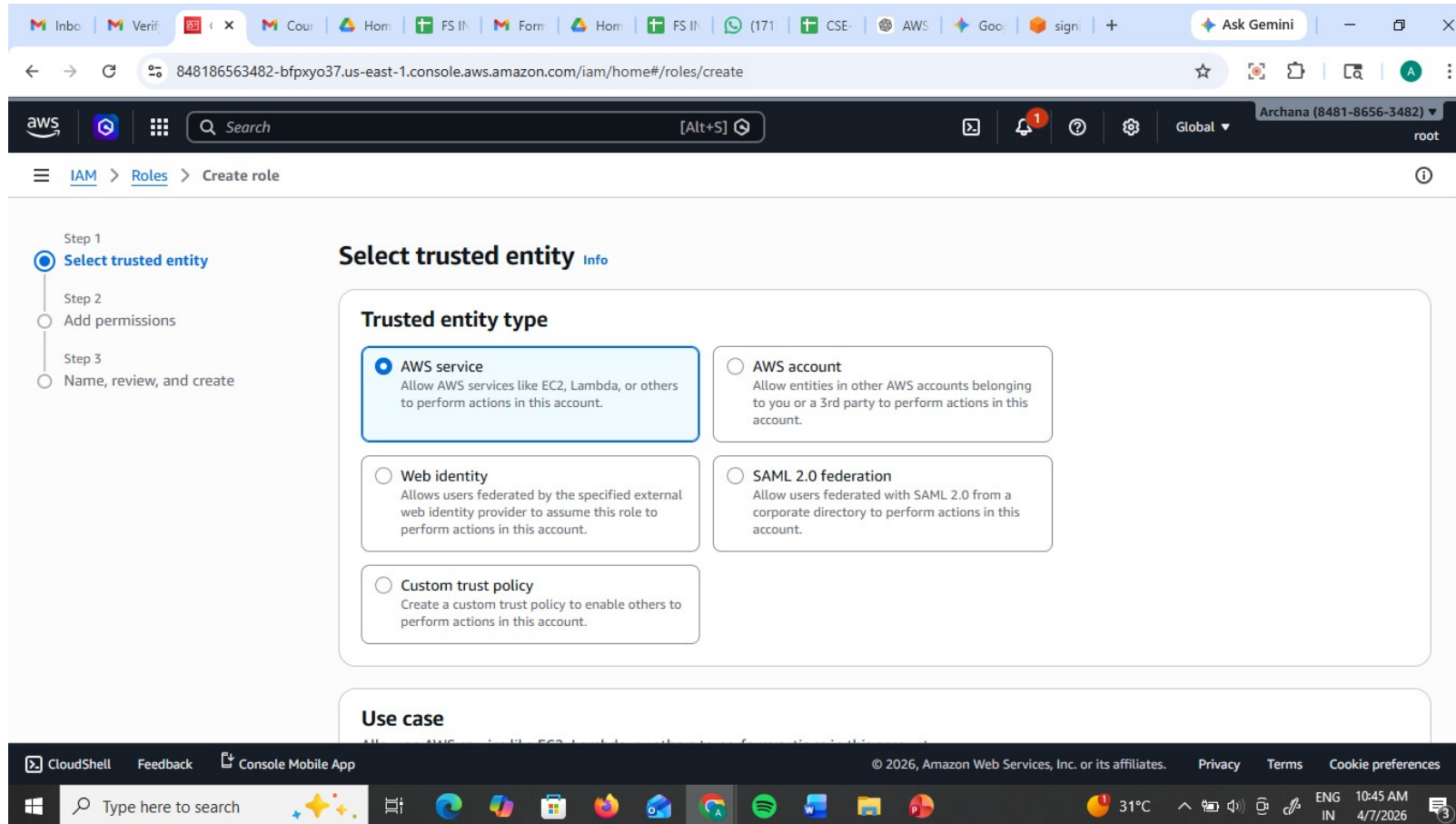
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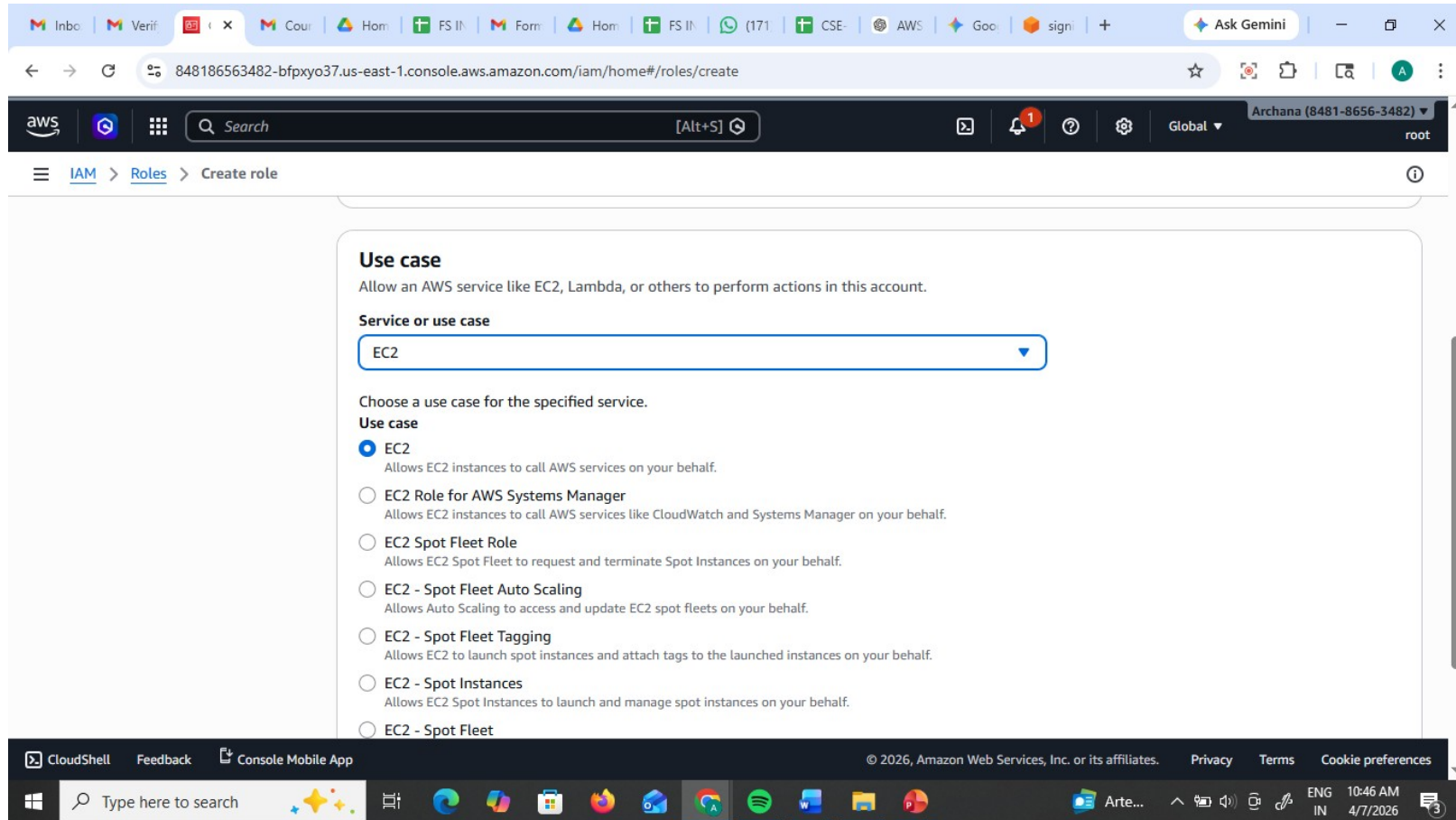
Type here to search

31°C 10:45 AM 4/7/2026

Select Trusted Entity: Choose AWS Service.



Under the "Service or use case" dropdown, select EC2. (This tells AWS: "I trust the EC2 service to use this role").



Add Permissions: Search for and check AmazonS3FullAccess.

The screenshot shows the AWS IAM console interface during the 'Create role' process. The left sidebar indicates the current step is 'Add permissions'. The main area is titled 'Add permissions' and shows a search for 'S3' with 26 matches. The 'AmazonS3FullAccess' policy is selected, highlighted in blue.

Permissions policies (1/1169)

Choose one or more policies to attach to your new role.

Filter by Type: All types (26 matches)

	Policy name	Type	Description
☐	AmazonDMSRedshiftS3Role	AWS managed	Provides access to manage S3 settings ...
☐	AmazonS3ExpressFullAccess	AWS managed	Provides full access to all S3 directory ...
☐	AmazonS3ExpressReadOnlyAccess	AWS managed	Provides read only access to S3Express...
☒	AmazonS3FullAccess	AWS managed	Provides full access to all buckets via t...
☐	AmazonS3ObjectLambdaExecutionRolePolicy	AWS managed	Provides AWS Lambda functions permi...
☐	AmazonS3OutpostsFullAccess	AWS managed	Provides full access to Amazon S3 on ...
☐	AmazonS3OutpostsReadOnlyAccess	AWS managed	Provides read only access to Amazon S...
☐	AmazonS3ReadOnlyAccess	AWS managed	Provides read only access to all bucket...

Name & Review: Name it (e.g., EC2-S3-ReadOnly-Role) and click Create role.

The screenshot shows the AWS IAM console interface for creating a new role. The browser address bar indicates the URL: `848186563482-bfpxyo37.us-east-1.console.aws.amazon.com/iam/home#/roles/create?trustedEntityType=AWS_SERVICE&selectedService=EC2&s...`. The AWS console header shows the user 'Archana (8481-8656-3482)' with 'root' permissions. The navigation pane on the left shows the path: IAM > Roles > Create role. The main content area is titled 'Name, review, and create' and contains the following sections:

- Step 1: Select trusted entities** (indicated by a blue circle in the left sidebar).
- Role details**:
 - Role name**: A text input field containing 'CC-Role-Demq'. Below the field, it states: 'Maximum 64 characters. Use alphanumeric and '+,_,@,-,_' characters.'
 - Description**: A text area containing 'Allows EC2 instances to call AWS services on your behalf.' Below the area, it states: 'Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+,-,@-/\[\]\#\\$\%^&*~'.'
- Trust policy**: A code editor showing a JSON snippet:

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": [
```

The bottom of the screen shows the Windows taskbar with various application icons and the system clock displaying '10:47 AM 4/7/2026'.

Name & Review: Name it (e.g., EC2-S3-ReadOnly-Role) and click Create role.

The screenshot shows the AWS IAM console 'Create role' page. The browser address bar shows the URL: `848186563482-bfpxyo37.us-east-1.console.aws.amazon.com/iam/home#/roles/create?trustedEntityType=AWS_SERVICE&selectedService=EC2&s...`. The AWS console header shows the user 'Archana (8481-8656-3482)' with 'root' permissions. The breadcrumb navigation is 'IAM > Roles > Create role'. The page is divided into three steps: Step 1: Name the role (partially visible), Step 2: Add permissions, and Step 3: Add tags. Step 2 is the active step, showing a 'Permissions policy summary' table with one entry: 'AmazonS3FullAccess' (AWS managed) attached as a 'Permissions policy'. Step 3 shows 'Add tags - optional' with a note that tags are key-value pairs and a button to 'Add new tag'. At the bottom right, there are 'Cancel', 'Previous', and 'Create role' buttons. The footer includes 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information for Amazon Web Services, Inc. or its affiliates.

Step 2: Add permissions

Permissions policy summary

Policy name	Type	Attached as
AmazonS3FullAccess	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional [Info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create role](#)

Role has been created

The screenshot displays the AWS IAM console interface. At the top, a green banner confirms 'Role CC-Role-Demo created.' with a 'View role' button. The main section, titled 'Roles (97)', includes a search bar with 'CC' entered, showing '3 matches'. A table lists the roles:

☐	Role name	Trusted entities	Last activity
☐	CC-Role-Demo	AWS Service: ec2	-
☐	dynamodbfull--access	AWS Service: lambda	17 days ago
☐	s3--full--access	AWS Service: lambda	706 days ago

Below the table, the 'Roles Anywhere' section is visible, featuring three cards: 'Access AWS from your non AWS workloads', 'X.509 Standard', and 'Temporary credentials'. The left sidebar shows the 'IAM > Roles' navigation path and a list of IAM-related options. The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 10:48 AM on 4/7/2026.

Create the ec2 instance and and connect to it and check for aws
ec2 describe-instances

```
ec2-user@ip-172-31-30-145:~  
ws.com  
The authenticity of host 'ec2-18-216-91-107.us-east-2.compute.amazonaws.com (18.216.91.107)' can't be e  
stablished.  
ED25519 key fingerprint is SHA256:2tOuEDdpBn5L50D48AT85MnMRfbLgAfMAQ7lMzGiWYw.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'ec2-18-216-91-107.us-east-2.compute.amazonaws.com' (ED25519) to the list of  
known hosts.  
  
#_  
~\_##### Amazon Linux 2023  
~~\_#####\  
~~\_####|  
~~\_#/_____  
~~\_V~'-'->  
~~~~  
~~._.  
__/_/  
_/m/'_/_
```

```
[ec2-user@ip-172-31-30-145 ~]$ aws ec2 describe-instances  
  
Unable to locate credentials. You can configure credentials by running "aws login".  
[ec2-user@ip-172-31-30-145 ~]$ aws s3 ls  
  
Unable to locate credentials. You can configure credentials by running "aws login".  
[ec2-user@ip-172-31-30-145 ~]$
```

Attach to Instance:
Go to the EC2 Dashboard.
Select your running instance.
Click Actions > Security > Modify IAM role.

The screenshot displays the AWS Management Console interface for the EC2 service. The top navigation bar shows the AWS logo, a search bar, and the current region (United States (Ohio)). The left sidebar contains the navigation menu with 'EC2' selected, leading to the 'Instances' page. The main content area shows a list of instances with one instance, 'Role-demo', selected. The instance details for 'i-0b711793773d1361b' are displayed below the list. The 'Actions' menu is open, showing options like 'Instance diagnostics', 'Instance settings', 'Networking', 'Security', 'Image and templates', 'Storage', and 'Monitor and troubleshoot'. The 'Security' option is highlighted, and its sub-menu is visible, showing 'Change security groups', 'Modify IAM role', and 'Get Windows password'. The 'Modify IAM role' option is selected.

EC2 > Instances

Instances (1/1) Info

Last updated 6 minutes ago

Connect Instance state Actions Launch instances

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

☒	Name	Instance ID	Instance state
☒	Role-demo	i-0b711793773d1361b	Running

i-0b711793773d1361b (Role-demo)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0b711793773d1361b	Public IPv4 address 18.216.91.107 open address	Private IPv4 addresses 172.31.30.145
IPv6 address	Instance state Running	Public DNS

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Select the role you just created and click Update IAM role

The screenshot shows the AWS Management Console interface in a web browser. The browser's address bar displays the URL: `848186563482-bfpxyo37.us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#ModifyIAMRole:instanceId=i-0b711793773d1361b`. The console header includes the AWS logo, a search bar, and navigation links for EC2, Instances, and the specific instance `i-0b711793773d1361b`. The main heading is "Modify IAM role" with an "Info" link. Below this, it says "Attach an IAM role to your instance." The "Instance ID" section shows `i-0b711793773d1361b` (Role-demo). The "IAM role" section has a dropdown menu currently set to "CC-Role-Demo" and a "Create new IAM role" button. At the bottom right of the main content area are "Cancel" and "Update IAM role" buttons. The footer of the console shows "CloudShell", "Feedback", "Console Mobile App", and copyright information for Amazon Web Services, Inc. The Windows taskbar at the very bottom shows the search bar, task view button, and several application icons.

AWS x Verify x Amaz x Inbo x Inbo x Home x (165) x Modi x EC2 I x AWS x + Ask Gemini

848186563482-bfpxyo37.us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#ModifyIAMRole:instanceId=i-0b711793773d1361b

aws [Search] [Alt+S] United States (Ohio) Archana (8481-8656-3482) root

EC2 > Instances > i-0b711793773d1361b > Modify IAM role

Modify IAM role [Info](#)

Attach an IAM role to your instance.

Instance ID
`i-0b711793773d1361b` (Role-demo)

IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

CC-Role-Demo [Create new IAM role](#)

[Cancel](#) [Update IAM role](#)

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Type here to search 32°C 8:01 PM 4/7/2026

Verify via CLI:
SSH/Connect into your EC2 instance.
Type `aws s3 ls`.
The Result: The command works immediately.

```
Select ec2-user@ip-172-31-30-145:~  
ED25519 key fingerprint is SHA256:2tOuEDdpBn5L50D48AT85MnMRfbLgAfMAQ7lMzGiWYw.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'ec2-18-216-91-107.us-east-2.compute.amazonaws.com' (ED25519) to the list of  
known hosts.  
  
#_  
~\####_ Amazon Linux 2023  
~~\#####\  
~~\###|  
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~V~'-'->  
~~~~  
~~._.  
_/_/_/  
_/m/'
```

```
[ec2-user@ip-172-31-30-145 ~]$ aws ec2 describe-instances  
  
Unable to locate credentials. You can configure credentials by running "aws login".  
[ec2-user@ip-172-31-30-145 ~]$ aws s3 ls  
  
Unable to locate credentials. You can configure credentials by running "aws login".  
[ec2-user@ip-172-31-30-145 ~]$ aws s3 ls  
2026-03-20 07:26:19 elasticbeanstalk-us-east-1-848186563482  
2026-03-25 06:43:38 iam-user-cc-lab-bucket-demo  
[ec2-user@ip-172-31-30-145 ~]$
```

We can here role has only s3 full access only here-- aws ec2 describe-instances – unauthorized

You did not have to provide an Access Key or Secret Key. AWS automatically provided Temporary Security Credentials to the instance.

```
Select ec2-user@ip-172-31-30-145:~
~\_####_ Amazon Linux 2023
~~\_#####\
~~\###|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~'-'>
~~~
~~~.~.
~~/_/
~/m/'

[ec2-user@ip-172-31-30-145 ~]$ aws ec2 describe-instances

Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-172-31-30-145 ~]$ aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-172-31-30-145 ~]$ aws s3 ls
2026-03-20 07:26:19 elasticbeanstalk-us-east-1-848186563482
2026-03-25 06:43:38 iam-user-cc-lab-bucket-demo
[ec2-user@ip-172-31-30-145 ~]$ aws ec2 describe-instances

An error occurred (UnauthorizedOperation) when calling the DescribeInstances operation: You are not authorized to perform this operation. User: arn:aws:sts::848186563482:assumed-role/CC-Role-Demo/i-0b711793773d1361b is not authorized to perform: ec2:DescribeInstances because no identity-based policy allows the ec2:DescribeInstances action
[ec2-user@ip-172-31-30-145 ~]$
```


Scenario based questions:

1. Scenario: New Developer Access

You hired a new developer who needs access to EC2 and S3.

Question: How will you provide access?

2. Scenario: Temporary Access Needed

A user needs access only for 2 days. How to do this?

3. Scenario: Access Key Leak

A developer accidentally exposed access keys on GitHub. How to resolve this?

4. Scenario: CLI Access Required

User needs programmatic access. How to do this?

5. Scenario: 50 Employees Same Permissions

50 employees need same S3 access. How to do this?

6. Scenario: Change Permissions for Multiple Users

You need to revoke access from 20 users. How to do this?

7. Scenario: Different Departments

HR, Dev, Admin need different access. How to do this?

8. Scenario: EC2 Access to S3

An EC2 instance needs access to S3 without credentials. How to do this?

9. Scenario: Cross-Account Access

Account A wants to access resources in Account B. How to do this?

10. Scenario: Lambda Access to DynamoDB

Lambda needs to read/write DynamoDB. How to give this?

11. Scenario: Temporary Elevated Access

Admin access needed temporarily. How to do this?

12. Scenario: Restrict S3 Access to Specific Bucket

User should only access one bucket. How to do this?

13. Scenario: Deny Delete Actions

Users should not delete resources. Why this happens?

14. Scenario: Time-Based Access

User should access only during office hours. How to do this?

15. Scenario: IP-Based Access

Allow access only from office network. How to do?

16. Scenario: Secure Best Practice

How do you manage permissions efficiently?

17. Scenario: Root User Usage

When should root user be used?

18. Scenario: Multiple Services Access

User needs EC2 + S3 + DynamoDB. How to do this?

19. Scenario: Audit Permissions

How to track user activity?

20. Scenario: Over-Permission Issue

User has too much permission. How to solve this?